Chapter 9

Knowledge Management and E-Learning

Introduction to Information Systems

Knowledge People Technology Learning

Presented by
Sundar Raj Sharma, TA
Dr. Angello Astorga, Instructor



What we will cover

Five learning objectives, in order.



9.1

Intellectual capital

What a company knows



9.2

Knowledge management

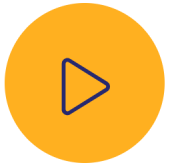
The steps to launch it



9.3

The human element

Why people hold back



9.4

E-learning

Different approaches



9.5

Creating e-learning

Teams, tools, and systems

Intellectual capital

What a company knows and who it knows. It adds value, but it is not on the financial reports.



Human capital

The skills and knowledge of the employees



Social capital

The relationships with customers, suppliers, and others



Structural capital

Documented processes, policies, contracts, and patents

Why it matters: it is often what sets a successful company apart from its competitors.

Two types of knowledge

Both types add to a company's intellectual capital.

Explicit

Knowledge that can be written down

Manuals, spreadsheets, websites, policies, and procedures

Tacit

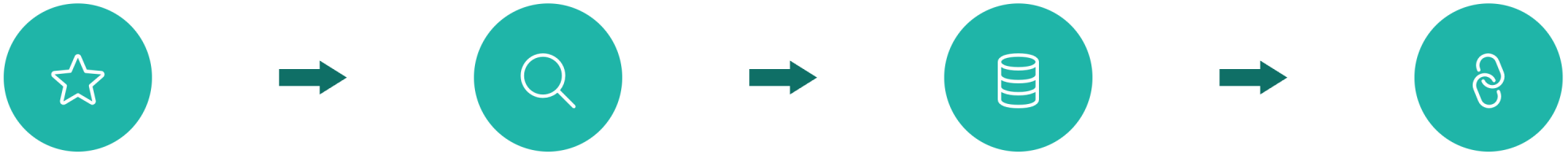
Knowledge that lives in people's heads

Insight, judgment, creativity, and wisdom from experience

The challenge: explicit knowledge is easy to store. Tacit knowledge is hard to capture.

Knowledge management in four steps

Knowledge management (KM): strategies and practices to manage a company's intellectual capital.



Step 1

Identify the goal

Pick one clear, specific aim

Step 2

Locate the sources

Find the information and the experts

Step 3

Capture the knowledge

Collect it in one place

Step 4

Organize, share, value

Make it easy to find and judge

Start here: a project with a specific goal is more likely to succeed. The goal also decides the methods and technology you use later.

Step 2: Locate the sources

Where is the knowledge, and who has it?

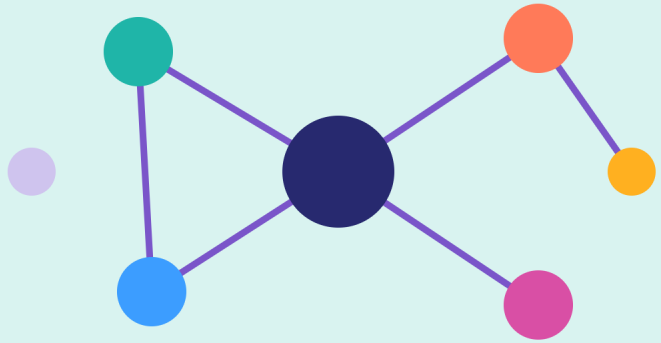
Find the experts

Expert directories
Online profiles list each person's projects and skills

Search software
Crawls databases, websites, email, and documents

Scattered files
Explicit knowledge is often already digital, but in many formats

Social network analysis



Dots are people, lines are relationships. It shows leaders, connectors, and loners.

Steps 3 and 4: Capture, organize, share

Collect the knowledge first. Then make it easy to find.

Capture

Knowledge base

Gathers explicit knowledge in one place

Open sharing

For tacit knowledge, let people share in their own words

Communities of practice

Groups that learn from each other using forums, wikis, and blogs

Organize and share

Enterprise content management

Controls documents, videos, web pages, and images

Document management

OCR reads paper forms and makes them searchable

The intranet

The company's own site, where people find and share

Step 4: Value the knowledge

Not everything captured is useful. The more valuable it is, the more effort it deserves.

Strategic information	Experiment with it and invest in it
Operational information	Collect and organize it, and share it widely
Compliance information	Automate collecting and archiving
Low-value or duplicate	Delete it

Tip: some companies let users rate entries in the knowledge base to judge their value.

Example: Mayo Clinic

Medical knowledge grows fast. Doctors need current, trusted answers when they treat a patient.



Experts contribute

Specialists share what they know

Editors refine

Short, practical “knowledge bytes”

Panels vet

Experts check the quality

Stored and shared

Organized with tags in one system

AskMayoExpert

Short, trusted answers to clinical questions

MayoExpert Decision Advisor

Combines patient data with clinical knowledge

The human element

Managers want everyone to share knowledge. Employees may see it differently.



Self-interest

What a person knows helps decide their salary, title, and promotions. Sharing it can feel like giving that away.



Hoarding

Keeping knowledge to stay valuable



Free riding

Using the knowledge base, never adding to it



No time

Sharing takes time from other work



Fear of criticism

Being corrected in public discourages people

Encouraging people to share

Incentives: rewards that influence behavior. They help, but they can backfire.

What works

- Show that sharing is valued
- Give moderate rewards
- Count sharing in performance reviews
- Reward team results, not only individuals

Watch out for

- Big cash bonuses: people may not share unless paid
- Gaming the system: tiny or split contributions
- Unfairness, when managers judge value differently

Tips for launching a KM project

1. Set a clear, specific goal and start small
2. Get management buy-in
3. Find the assets and experts to fill the knowledge base with accurate, current content
4. Choose simple technology that fits existing systems
5. Run a pilot with a small group of willing employees
6. Plan incentives that fit your organization
7. Encourage people to take part and suggest improvements

Start small, then grow

Three approaches to e-learning

E-learning: learning that uses the internet to connect learners with materials, teachers, and each other.



Self-paced

Learn alone, anytime, with little or no teacher. Good for company policies and systems.



Instructor-led

A teacher guides the course using virtual classrooms, web conferencing, email, and forums.



Hybrid

Online activities blended with in-class sessions.

Synchronous: everyone at the same time, from different places.

Asynchronous: each person at any time they choose.

E-learning in universities and companies

Both use e-learning, but they started in different ways.

MOOCs

Massive open online courses

Open to anyone, free or low cost, often through a third party like edX.

Millions enroll, but most never finish.

In companies

Self-paced modules for skills and certifications

Example: Google's Data Analytics courses lead to a professional certificate.

The difference: universities tried to copy the face-to-face class. Companies built self-paced modules.

Who creates an e-learning course?

Objective 9.5: it takes a team of four roles.



Subject matter expert

Knows the content



Instructional designer

Plans how to teach it and how to test it



Project sponsor

Sets the goals and pays the bills



Project manager

Coordinates the work and tracks progress

Building the course content

Learning objects: small, reusable pieces of course content.

Narrated slides

A presentation with a voice explaining it

Simulations

Choices lead to consequences, with no real risk

Serious games

Games built to teach, not just entertain

AR and VR

Learn inside a realistic, immersive scene

Content is king. Substance comes first, technology second.

Preventing cheating online

Online, no one is in the room, so schools use three kinds of safeguards.



Identity

Is it really you? Passwords, fingerprints, and other biometrics.



Honesty

Honor codes, proctored tests, and webcam room scans.



The AI challenge

AI-detection tools help. Eye tracking, browser monitoring, and typing patterns are in development.

Learning management systems

LMS: a system that delivers courses, tracks progress, and keeps records.

Deliver

Serves up courses to learners

Track

Records progress and results

Learning analytics

Uses LMS data to show how students are doing

Social learning

Wikis, blogs, forums, and virtual classrooms

SCORM: a standard that lets learning objects work in any LMS that supports it.

Online vs. classroom learning

Two questions: same place or not, and same time or not.

DIFFERENT PLACE · SAME TIME

Synchronous online

Live video, virtual classrooms, chat, webcams, whiteboards

DIFFERENT PLACE · DIFFERENT TIME

Asynchronous online

Forums, email, wikis, blogs, video on demand

SAME PLACE · SAME TIME

Classroom

Lectures, smartboards, laptops, audience response

SAME PLACE · DIFFERENT TIME

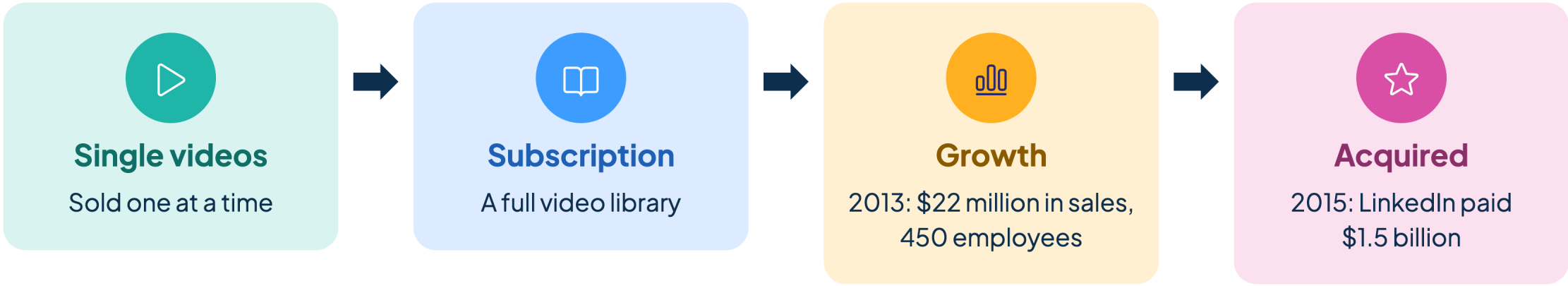
Open labs

Computer labs, language labs

Research finds few differences. The two complement each other, so blend both.

Case study: Lynda.com

An online video-training company that grew until it was bought twice over.



Then Microsoft bought LinkedIn for $25.2 billion. Next: videos plus LinkedIn's network plus Microsoft's analytics and AI.

Key takeaways

9.1

Human, social, and structural capital. Explicit and tacit knowledge.

9.2

Four steps to launch KM: goals, locate, capture, organize.

9.3

People hold back. Fair, moderate incentives help.

9.4

Self-paced, instructor-led, and hybrid e-learning.

9.5

A team builds courses; an LMS delivers and tracks them.

Knowledge and learning are what make people and organizations grow.